

Customer Churn Dashboard Report

1. Project Overview

This dashboard was developed in **Power BI** to analyze customer churn behavior and identify key factors associated with customer attrition. The dashboard provides an interactive view of churn trends, service usage patterns, and geographic distribution, help users explore the data through filters and slicers.

2. Data Preparation

The dataset was imported into Power BI and prepared for analysis. The preparation process included:

- Reviewing and validating data types
- Creating calculated measures such as:
 - **Total Customers**
 - **Churned Customers**
 - **Retained Customers**
 - **Churn Rate**
 - **Average Usage Metrics**
- Creating categorical labels for:
 - **Churn Label**
 - **International Plan Label**
 - **Voice Mail Plan Label**
- Organizing the data model to support interactive dashboard analysis

3. Dashboard Structure

The report consists of two main pages:

A. Customer Churn Overview

This page provides a high-level summary of customer churn, including:

- Total number of customers

- Number of customers churned
- Number of customers retained
- Overall churn rate
- Churn rate by customer service calls
- Churn rate by international plan
- Customer distribution by churn label
- Geographic churn distribution by state

B. Usage Analysis

This page focuses on customer usage behavior and compares churned vs. retained customers in terms of:

- Average day minutes
- Average evening minutes
- Average night minutes
- Average international minutes
- Average total charge

4. Interactive Features

The dashboard includes slicers and filters that allow users to explore the data dynamically by:

- **Voice Mail Plan**
- **International Plan**
- **State**

These interactive controls help users drill into specific customer segments and compare churn behavior across different groups.

5. Key Findings

1) Overall Churn Rate

The dashboard shows:

- **Total Customers:** 2,666
- **Churned Customers:** 388
- **Retained Customers:** 2,278
- **Churn Rate:** 14.6%

This indicates that a notable portion of customers left the service, making churn reduction an important business priority.

2) Customer Service Calls Have a Strong Relationship with Churn

One of the clearest insights is that churn rate rises significantly as the number of **customer service calls** increases.

- Customers with **0–3 service calls** have relatively low churn
- Starting from **4 calls**, churn rises sharply
- At **7–9 service calls**, churn becomes extremely high

Interpretation:

Frequent service calls may indicate unresolved issues, dissatisfaction, or poor customer support experience. This suggests that customers who repeatedly contact support are at much higher risk of leaving.

3) International Plan Customers Are More Likely to Churn

The dashboard shows that customers with an **International Plan** have a much higher churn rate than those without it.

Interpretation:

This may indicate pricing concerns, low satisfaction with plan value, or unmet expectations among international plan users. This segment should be monitored more closely.

4) Churned Customers Show Higher Usage

From the Usage Analysis page, churned customers generally have higher averages across all major usage metrics:

- **Avg Day Minutes:** higher for churned customers
- **Avg Evening Minutes:** higher for churned customers
- **Avg Night Minutes:** slightly higher for churned customers

- **Avg International Minutes:** higher for churned customers

Interpretation:

Customers with higher usage may also have higher expectations and higher bills, making them more sensitive to service quality or pricing issues.

5) Churned Customers Have Higher Average Total Charges

The dashboard indicates that customers who churned had a higher **average total charge** compared to retained customers.

Interpretation:

Higher charges may contribute to churn, especially if customers feel they are not receiving enough value for the amount they pay.

6) Geographic Differences Exist Across States

The state-level map suggests that churn varies geographically, meaning some states may experience higher churn than others.

Interpretation:

This could be useful for regional targeting, localized retention strategies, or identifying areas where service or competition may be affecting customer loyalty.

6. Business Recommendations

Based on the dashboard insights, the following actions are recommended:

- 1. Monitor customers with high customer service call frequency**
 - Customers with 4 or more service calls should be flagged as high-risk churn candidates
- 2. Improve issue resolution in customer support**
 - Reducing repeated service calls may help lower churn
- 3. Review the International Plan offering**
 - Reassess pricing, features, and customer value proposition for international plan users
- 4. Target high-usage customers with retention strategies**
 - Offer loyalty incentives, personalized plans, or pricing support to customers with high usage and charges
- 5. Investigate high-churn states**

- Use geographic analysis to identify where churn is highest and apply region-specific actions

7. Conclusion

The Power BI dashboard successfully highlights the main drivers and patterns of customer churn. The analysis shows that **customer service interactions, international plan subscription, and higher usage/charges** are strongly associated with churn. By using these insights, businesses can take proactive actions to improve customer retention and reduce churn risk.